16 · Revision set: one question per idea, for the night before

Answer from memory on paper, then check the answers page. Any miss points back to its chapter, shown in brackets.

Draw from memory

1. The zener regulator (Exp 2) with values, the clamper (Exp 4), and the CE amplifier (Exp 6) with every part labelled. Then say each part's job in one line.

Diodes

2. A DSO shows 6 divisions peak-to-peak at 0.5 V/div on a sine. Give V_{rms}. *(Exp 0)*
3. Zener: define dynamic resistance, and say why "dynamic". *(Exp 1)*
4. A 6.2 V zener: which breakdown mechanism, and why is its temperature drift small? *(Exp 1)*
5. In the zener regulator, V_{in} goes up by 3 V. Which part takes almost all of it? *(Exp 2)*
6. Write the formulas for % line regulation and % load regulation. *(Exp 2)*
7. A positive clipper uses a 3 V battery. Clipping level? *(Exp 3)*
8. Clipper vs clamper: what happens to V_{pp} in each? *(Exps 3–4)*
9. What condition on RC makes a clamper work at 1 kHz? *(Exp 4)*

BJT

10. Bias of the two junctions in active, saturation and cut-off. *(Exp 5)*
11. Name the four h-parameters, and say from which curve, with what held constant, each is read. *(Exp 5)*
12. In Exp 6, which capacitor mostly sets f_L, and why does the gain fall at high f? *(Exp 6)*
13. Define bandwidth and the −3 dB point. *(Exp 6)*
14. Why is the CE output 180° out of phase? *(Exp 6)*
15. Emitter follower: gain, phase, input and output resistance (high/low). *(Exp 7)*
16. Removing C_E: name the type of feedback, and two things that improve. *(Exp 8)*

FET, oscillator, MOSFET

17. Define g_m, r_d, μ and show $\mu = g_m r_d$. *(Exp 9)*
18. Write Shockley's equation, and define I_{DSS} and V_P. *(Exp 9)*
19. CS amplifier gain formula, and why the gate divider can use megohms. *(Exp 10)*
20. Barkhausen criterion, and the frequency formula of the RC phase-shift oscillator. *(Exp 11)*
21. Why must the amplifier's gain be at least 29 in the phase-shift oscillator? *(Exp 11)*
22. Resistive-load inverter: the MOSFET's state and the output for each input. Define V_M. *(Exp 12)*
23. Two cascaded stages: what happens to the gain and to the bandwidth? *(Extra)*
24. In LTspice, how do you enter 1 MΩ, and which analysis gives a frequency response? *(01)*

Answers

1. See each chapter's redrawn figure and part table.
2. V_{pp} = 3 V, so V_p = 1.5 V and V_{rms} = 1.5/√2 = **1.06 V**.
3. $r_z = \Delta V_Z/\Delta I_Z$ in breakdown: the slope of the curve, not V/I at one point.
4. A mix of tunnelling (Zener) and avalanche. Their temperature coefficients are opposite (− and +), so they nearly cancel.
5. The series resistor R1. The extra current goes into the zener, and V_o barely moves.
6. Line: $\Delta V_o/\Delta V_{in} \times 100$. Load: $(V_{NL} - V_{FL})/V_{FL} \times 100$.
7. 3 + 0.7 = **3.7 V**.
8. A clipper reduces V_{pp}. A clamper keeps V_{pp} and shifts the DC level.
9. RC ≫ T (= 1 ms). Here RC = 10 ms.
10. Active: BE forward, BC reverse. Saturation: both forward. Cut-off: both reverse.
11. $h_{ie} = \Delta V_{BE}/\Delta I_B$ (I_B fixed). $h_{re} = \Delta V_{BE}/\Delta V_{CE}$ (I_B fixed). $h_{fe} = \Delta I_C/\Delta I_B$ (V_{CE} fixed). $h_{oe} = \Delta I_C/\Delta V_{CE}$ (I_B fixed).
12. The emitter bypass C_E. At high f, the transistor's internal junction capacitances (Miller effect) and stray capacitance shunt the signal.
13. BW = $f_H - f_L$, between the frequencies where the gain is 0.707 × max = −3 dB (half power).
14. A rising base voltage raises I_C, which increases the drop across R_C, so the collector voltage falls.
15. Gain ≈ 1 (slightly less), in phase; input resistance high, output resistance low; used as a buffer.
16. Current-series negative feedback. Gain becomes stable (independent of β), bandwidth widens, distortion and noise fall, input resistance rises.
17. $g_m = \Delta I_D/\Delta V_{GS}$ (V_{DS} const), $r_d = \Delta V_{DS}/\Delta I_D$ (V_{GS} const), $\mu = -\Delta V_{DS}/\Delta V_{GS}$ (I_D const) = $(\Delta V_{DS}/\Delta I_D) \times (\Delta I_D/\Delta V_{GS}) = r_d g_m$.
18. $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$. I_{DSS} = drain current at V_{GS} = 0 in saturation. V_P = the V_{GS} at which $I_D \to 0$.
19. $A_v = -g_m(R_D \parallel R_L)$. The gate draws no current, so megohm resistors drop no voltage.
20. $|A\beta| \geq 1$ and loop phase 360°. $f = 1/(2\pi RC\sqrt{6})$.
21. At the oscillation frequency, the three-section RC ladder attenuates the signal to 1/29.
22. Input low: MOSFET off, output = V_{DD}. Input high: MOSFET on, output ≈ 0. V_M = the input voltage at which $V_{out} = V_{in}$.
23. Gains multiply (dB add). Bandwidth shrinks.
24. `1Meg` (`1M` is milli). `.ac`.